STAT 579: Applied Causal Inference

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Instructor & Logistics

Instructor	Dr. Davood Tofighi
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Email	Through Canvas
Office Hours	TBD – Office/Zoom
Class Time	Tuesdays & Thursdays, 9:30 AM – 10:45 AM

Course Overview

This graduate-level seminar introduces the **theory and application of causal inference**, with a strong emphasis on **biostatistical and epidemiological contexts**. The course builds from **conceptual foundations and core assumptions** to **modern estimation methods** for causal effects, emphasizing **target trial emulation** as a unifying framework.

Key frameworks and tools include:

- Potential Outcomes Framework & SUTVA
- Directed Acyclic Graphs (DAGs)
- Propensity Scores, IPW, and Doubly Robust Methods
- Instrumental Variables
- Mediation Analysis (NDE/NIE)
- Marginal Structural Models (MSMs)
- Sensitivity Analysis

Instructor's Approach

Causal inference is not just a toolbox—it's a **rigorous way of thinking** about cause and effect. We will prioritize:

- **Precise causal questions** (not just statistical associations).
- **Transparency of assumptions** (via DAGs and formal notation).
- **Identification before estimation** (thinking carefully about what is learnable from data).
- **Critical robustness assessment** (sensitivity, replication, target trial framing).

Learning Objectives

By the end of the course, students will be able to:

1. Formulate and distinguish causal from associational questions.
2. Represent assumptions using DAGs and counterfactual notation.

3. Evaluate and apply key assumptions (SUTVA, exchangeability, positivity).
4. Implement modern causal inference methods (propensity scores, IPW, MSMs, IVs, G-computation).
5. Decompose effects via mediation analysis.
6. Conduct sensitivity analyses for unmeasured confounding.
7. Frame observational studies as target trial emulations.
8. Critically evaluate causal claims in the literature.

Assessment and Grading

Component	Weight	Description
Weekly Concept Checks	10%	Short, low-stakes exercises on assumptions, DAGs, and interpretation.
Problem Sets (3–4 sets)	25%	Analytical and coding assignments covering key methods (e.g., propensity scores, IV, mediation).
Paper Critique & Discussion Lead	20%	Each student will lead one class discussion of an assigned paper. This involves reconstructing the authors' DAG, explicitly stating their assumptions, and proposing a plan to assess the robustness of the findings.
Final Project	35%	Independent research project framed as a target trial emulation. Includes a 10% proposal and 25% final paper.
Final Presentation	10%	15-minute conference-style presentation defending your project's causal assumptions and interpreting the results.

Course Materials

Required (Freely Available Online)

- Hernán, M. A., & Robins, J. M. (2024). *Causal Inference: What If*. Chapman & Hall/CRC. https://www.hsph.harvard.edu/miguel-hernan/causal-inference-book/
- Neal, B. (2020). *Introduction to Causal Inference*. https://www.bradyneal.com/causal-inference-course

Recommended

- Huntington-Klein, N. (n.d.). *The Effect: An Introduction to Research Design and Causality | The Effect*. Retrieved August 22, 2025, from https://theeffectbook.net/
- Barrett, M., McGowan, L. D., & Gerke, T. (2025, August 21). Causal Inference in R. https://www.r-causal.org/
- Cunningham, S. (2021). *Causal Inference: The Mixtape*. Yale University Press. URL: https://mixtape.scunning.com
- VanderWeele, T. J. (2015). *Explanation in Causal Inference*. Oxford University Press, 2015. (Excellent resource specifically for mediation).
- Pearl J, Glymour M, Jewell NP. *Causal Inference in Statistics: A Primer.* Wiley, 2016. (Primer is a more accessible introduction to Pearl's framework). Focuses heavily on DAGs and structural causal models.

- Imbens GW, Rubin DB. *Causal Inference for Statistics, Social, and Biomedical Sciences: An Introduction.* Cambridge University Press, 2015. (Focuses heavily on the potential outcomes framework and design-based inference).

Software

- **Required:** R and RStudio (or another modern IDE like VS Code). While code will be demonstrated, students are expected to have intermediate proficiency in R. The focus will be on the application and interpretation of packages, not on learning R from scratch.
- **Core Packages:** `tidyverse`, `dagitty`, `ggdag`, `MatchIt`, `WeightIt`, `cobalt`, `ivreg`, `mediation`, `EValue`.

Course Schedule (14 Weeks)

Part I: Foundations

Week	Dates	Core Topic	Primary Readings	Project Link	Paper Critique
1	Aug 19, 21	What is a Causal Effect? Counterfactuals & SUTVA	HR Ch. 1–2; Neal Ch. 1–2	Define your causal question	Instructor-led demo critique (introducing assignment expectations)
2	Aug 26, 28	DAGs & Assumption Transparency	Neal Ch. 3; HR Ch. 6	Draw a preliminary DAG	Student critique
3	Sep 2, 4	Exchangeability, Positivity, Consistency	HR Ch. 7; Neal Ch. 4	Identify assumptions for your DAG	Student critique
4	Sep 9, 11	Confounding, Selection, Measurement Bias	HR Ch. 8–9	Consider biases in your dataset	Student critique

Part II: Estimation Methods

Week	Dates	Core Topic	Primary Readings	Project Link	Paper Critique
5	Sep 16, 18	Propensity Scores (matching, weighting)	HR Ch. 12–13; Neal Ch. 7	Select adjustment strategy	Student critique
6	Sep 23, 25	IPTW, MSMs, Doubly Robust Estimators	HR Ch. 15; Neal Ch. 7.5–7.6	Evaluate MSM relevance	Student critique
7	Sep 30, Oct 2	G-computation	HR Ch. 20–21	Explore g-methods for your project	Student critique
8	Oct 7	Instrumental Variables	HR Ch. 16; Neal Ch. 9	Assess natural experiments	Student critique

Note: *UNM Fall Break is Oct 9–10. Thursday's class is cancelled.*

Part III: Advanced Topics & Applications

Week	Dates	Core Topic	Primary Readings	Project Link	Paper Critique
9	Oct 14, 16	Effect Modification	HR Ch. 4–5; Neal Ch. 10	**Project Proposal Due**	Student critique
10	Oct 21, 23	Mediation Analysis I (Foundations)	VanderWeele Ch. 1–2; Neal Ch. 14	Refine DAG with mediators	Student critique
11	Oct 28, 30	Mediation II & Principal Stratification	VanderWeele Ch. 3; HR Ch. 23	-	Student critique
12	Nov 4, 6	Sensitivity Analysis	Neal Ch. 8; HR Ch. 14	Plan robustness checks	Student critique
13	Nov 11, 13	Target Trial Emulation	HR Ch. 22	Reframe project as trial	Student critique
14	Nov 18, 20	**Final Presentations**	–	Present findings	–

Note: *Thanksgiving Break is Nov 27-28.*

Finals Week (Dec 8–13): Final Project Paper Due.

Course Policies

- **Late Work:** Deadlines are firm. Late work will be penalized 10% per day unless a prior arrangement is made for a documented emergency.
- **Academic Integrity:** Collaboration on ideas is encouraged, but all submitted work must be your own.
- **Participation & Respect:** Active, inclusive, and evidence-based dialogue is expected.
- **Accessibility:** Students requiring accommodations should contact the Accessibility Resource Center and provide the instructor with an official letter.